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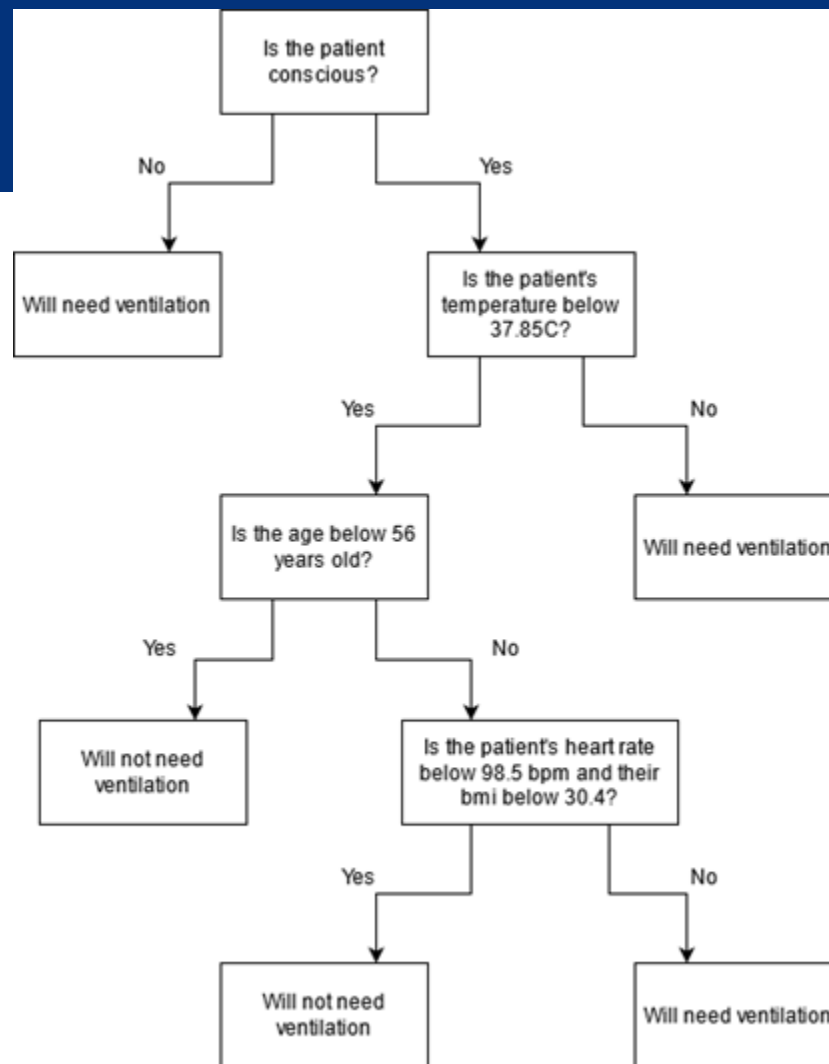
Supervised learning

A natural approach

Decision Trees

- ❖ The Queen before SVM...
- ❖ ... which was the Queen before DL

Decision Trees



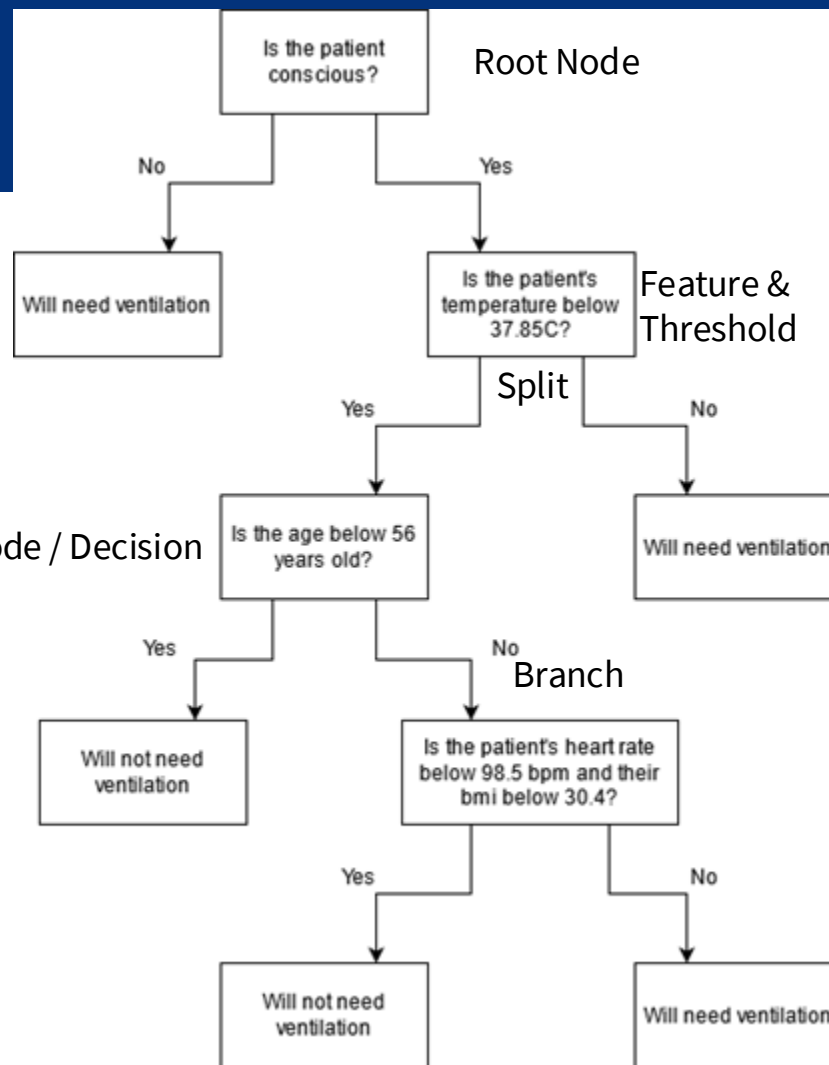
Decision Trees

Decision stump (depth 1)

Intermediate Node / Decision

Branch

Leaf / Terminal node



Decision Trees

- ❖ Ordered list of questions/tests (*if nest*)
- ❖ Recursive split until homogeneity (or stop condition)
 - Each branch gets us closer to purity
- ❖ Does not require normality (non-parametric)
- ❖ Linearity?

Decision Trees

- ❖ Ordered list of questions/tests (*if nest*)
- ❖ Recursive split until homogeneity (or stop condition)
 - Each branch gets us closer to purity
- ❖ Does not require normality (non-parametric)
- ❖ Non linear/Piecewise linear model

Decision Trees

- ❖ Search tree space
 - Ordered sequence of features
 - Splitting point
- ❖ aka CART

Decision Trees

❖ What feature?

- Categorical DT:

- Node impurity (Gini) or Information Gain (Entropy)

- Continuous DT:

- Node variance (MSE)

Decision Trees

❖ What splitting points?

■ Categorical vars:

- Test all
- Exhaustive

■ Continuous vars:

- Test mean of consecutive pairs in order
- Heuristic

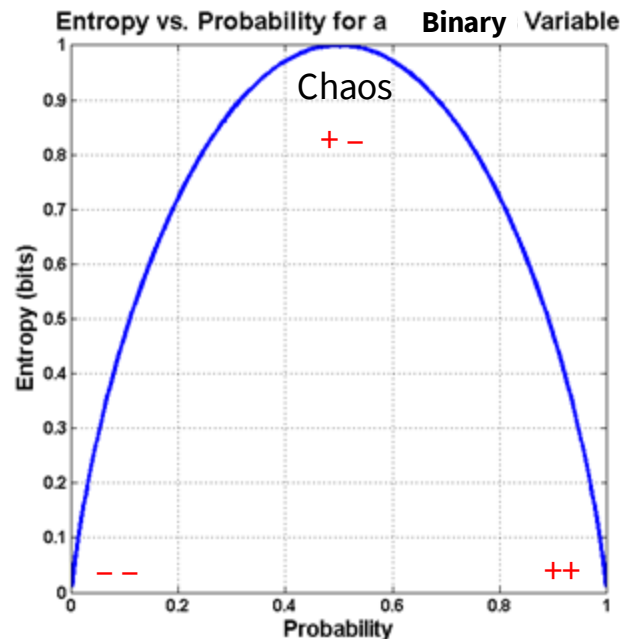
Entropy & Decision Trees

- ❖ feat. Information Theory

“entropy characterizes the purity of an arbitrary collection of examples”

- ❖ Information Gain:

- Entropy of parent - sum of child (branches) entropies
 - weighted by num. samples in child



Entropy & Decision Trees

- ❖ *Information Gain* biased towards feats w/ many distinct values
 - Identifiers have zero entropy! OF fest
 - *Gain Ratio* penalizes many outcomes
 - Intrinsic Information: entropy of the attribute's outcomes
 - Increases w/ num. branches
 - $GR = IG/II$
 - Implementations
 - ID3/C4.5/C5.0

Entropy & Decision Trees

❖ Gini impurity

- Prob. of misclassification given overall distribution

Percentage of samples of class j in current node

$$\sum p_j \underbrace{(1 - p_j)}_{\text{Prob. of misclassification in class j}} = \sum (p_j - p_j^2) = \left(\sum p_j\right) - \left(\sum p_j^2\right) = 1 - \sum p_j^2$$

Ins.	Class
1	A
2	A
3	B
4	B

Entropy & Decision Trees

❖ Gini impurity

- Prob. of misclassification given overall distribution

$$\sum p_j \underbrace{(1 - p_j)}_{\substack{\text{Percentage of samples of class } j \text{ in current node} \\ \text{Prob. of misclassification in class } j}} = \sum (p_j - p_j^2) = \left(\sum p_j\right) - \left(\sum p_j^2\right) = 1 - \sum p_j^2$$

❖ Weighted Gini (by num. samples in child)

Information gain, Gain ratio or Gini impurity?

- ❖ IG and Gini: Only in 2% of cases they are different
- ❖ Gini is fastest of all
- ❖ Gain ratio is slowest of all

Compute Gini impurity

❖ Donat un dels 7 llindars d'edat

❖ Calcula les branques

❖ La Gini impurity total

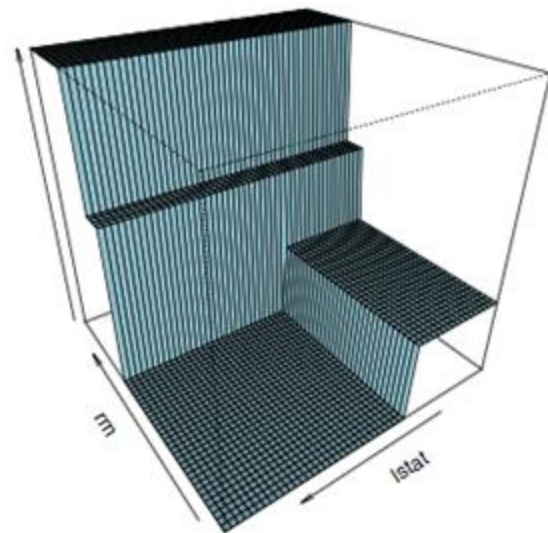
Edat	Compra/Lloguer
24	L
26	L
27	C
29	L
30	C
32	C
35	L
36	C

Decision Trees

- ❖ Hyperparameters (mostly to prevent OF)
 - Depth
 - Min. samples per node/leaf
 - Min. gain

Good Trees

- ❖ All vars welcome
 - Binary, discrete (prediction is mode), continuous
- ❖ Highly interpretable (if shallow)
- ❖ Feature relevance
 - This is important!
- ❖ High-dimensionality resistant (no distance!)
- ❖ A 5 min. baseline for everything

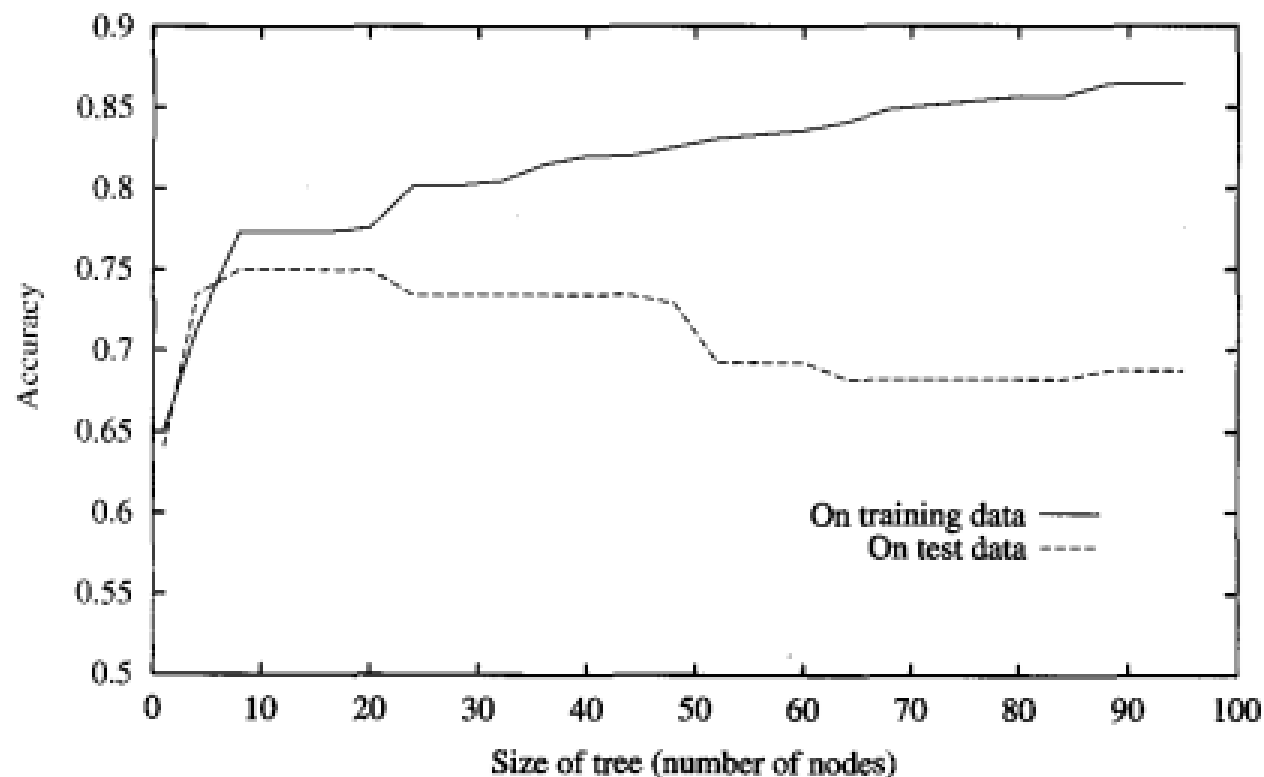


[24,25,26,27]

Bad Trees

- ❖ Buckets lacks finesse (piecewise linear)
- ❖ Greedy (cost grows with # feats.)
- ❖ Not great for regression (mean)
- ❖ Sensitive to small training sets (ratio classes/samples)
- ❖ Sensitive to class imbalances (gain weighted by samples in branch)
- ❖ Overfitting 100% guaranteed! (split until 1 node / 1 sample)

Overfitting Trees



Avoiding overfit

- ❖ Limits (pre-pruning)
- ❖ Backtracking (Post-pruning)
- ❖ Folds & Ensembles

Avoiding overfit

❖ Pre-pruning

- Min. samples for split
- Min. samples per leaf
- Min. gain
- Max. depth
 - Bias vs Variance
 - UF vs OF

Avoiding overfit

❖ Post-pruning

- Validation data. Prune subtrees that do not improve
 - Prediction at pruned node becomes mode/average of subtree
- Less hyperparams than pre-pruning.

Avoiding overfit

❖ Folds & Ensembles

- More hyperparams
 - Num. trees
 - Subsampled features
 - Subsampled samples
- Lost interpretability
- More expensive

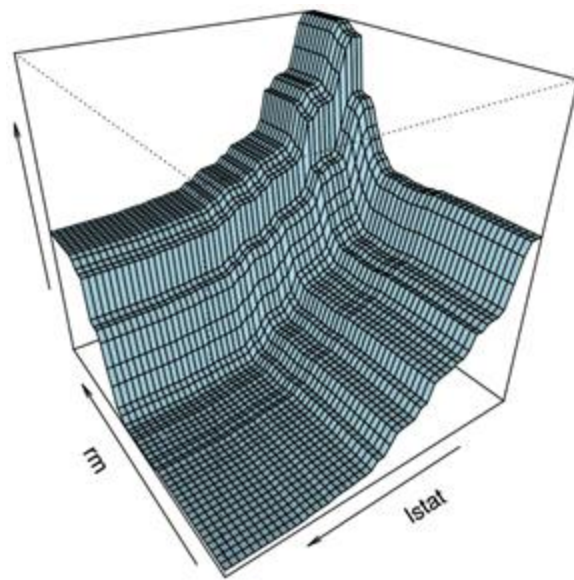
Putting trees together

❖ Random Forest

- Different data/features (bagging)
- Independent models (trained in par.)
- Aggregate outcome (mode or average)
- Noise resistant

❖ The price

- Simplicity
- Interpretability



Putting trees together

❖ XGBoost

- Same data/features (boosting)
- Dependent models (one after another)
- Reweighting of errors with gradient
- Guided by iterative loss minimization
- Highly efficient implementation
- Trees on steroids. Kaggle MVP. [\[40,41\]](#)

❖ The price

- Simplicity
- Interpretability

